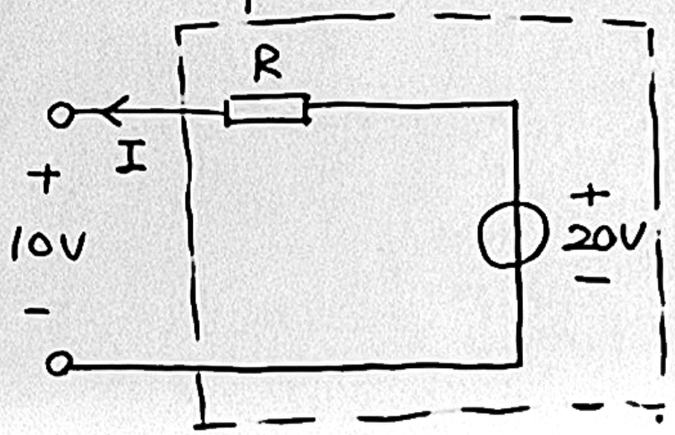
样题4详解

1.



KVL: $20 = IR + 10$.

$\therefore I = \frac{10}{R} > 0$.

考虑左例，关联参考方向. $P = UI > 0$，吸收.

则N是 发出

B.

2. 易求: $\varphi_a = 1V$, $\varphi_b = 3V$ $\therefore U = \varphi_a - \varphi_b = -2V$ D

3. 由Y→Δ: $R_\Delta = \frac{3 \cdot R_Y^2}{R_Y} = 3R_Y$ B.

4. 理想电压源内阻 $r = 0$ A.

5. $w \to 2w$, R不变, $jwL \to j \cdot 2wL$ $\therefore$ $1 + j4 \to 1 + j8$ D.

6. 左R_2与电流源串联，无效.

节点1: $U_{n1} \cdot \frac{1}{R1} = is - i$ B.

7. 网孔2： $(200+R)I_2 - 200I_1 = -rI_1$.

$\rightarrow (200+R)I_2 = (200-r)I_1$.

原： $400I_2 = 100I_1$.

$\therefore \frac{200+R}{400} = \frac{200-r}{100}$，理论上 r 是无穷多解的，但若要求

形式完全一致，只好 $r = 100\Omega$. A.

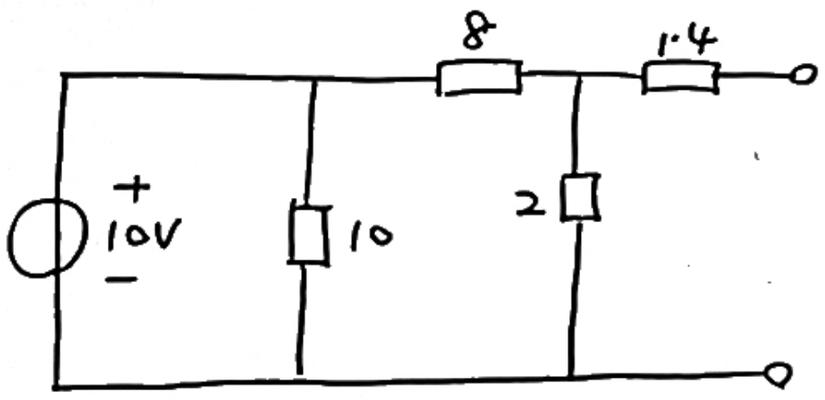
8. $\dot{i}_1 = -4\cos(100t+60^\circ) = 4\cos(100t-120^\circ)$.

$\dot{i}_2 = 4\sin(100t+60^\circ) = 4\sin(100t-30^\circ+90^\circ) = 4\cos(100t-30^\circ)$.

$\therefore \Delta\varphi = -120^\circ - (-30^\circ) = -90^\circ$ D.

（系数同号，函数同名）.

9.



戴维南： 求 R_{eq}： $8//2 + 1.4 = 3\Omega$.

求 U_{OC}： $10 \times \frac{2}{8+2} = 2V$.

$\therefore R_L = R_{eq} = 3\Omega$ B.

10. $\dot{i} = -\frac{1}{2}u + 2$.

$u = -iR_S + u_{OC}$. $\Rightarrow$. $\frac{u}{R_S} = -i + \frac{u_{OC}}{R_S}$ $\Rightarrow$ $i = -\frac{1}{R_S}\cdot u + \frac{u_{OC}}{R_S}$.

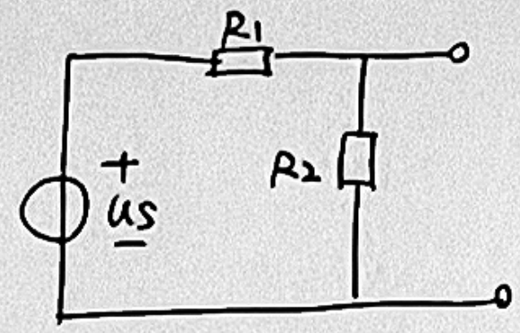
$\Rightarrow \begin{cases} -\frac{1}{R_S} = -\frac{1}{2} \\ \frac{u_{OC}}{R_S} = 2 \end{cases}$ $\therefore \begin{cases} R_S = 2 \\ u_{OC} = 4 \end{cases}$. B

11. 打开前，L与R3被短路，$i_L(0^-) = 0 = i_L(0^+)$.

打开后，R2串R3串L支路无电流，$u(0^+) = u_{R1} = 2 \times 20 = 40V$.

D.

12. 戴维南：



$U_{oc} = U_s \cdot \frac{10}{10+10} = 10\varepsilon(t)$

$R_{eq} = R_1 // R_2 = 5\Omega$.

$\tau = \frac{L}{R_{eq}} = \frac{1}{5}s$. $\Rightarrow$ **B**

13. 还原为标准相量：$\dot{U}_{sm} = 10\sqrt{2}\angle 0°$ V.

$\dot{I}_s = \frac{10\sqrt{2}\angle 0°}{10 - j10} = \frac{10\sqrt{2}\angle 0°}{10\sqrt{2}\angle -45°} = 1\angle 45°$ A.

$\dot{I} = 2\dot{I}_s' = 2\angle 45°$ A $\quad I = 2A$ **A**

14. 基波：$w = 50$，$j\omega L = j \cdot 50 \cdot 0.2 = j10$

$\dot{U}_{s1} = \frac{5}{\sqrt{2}}\angle 0°$，$\dot{I}_{s1} = \frac{\dot{U}_{s1}}{j10} = \frac{\frac{5}{\sqrt{2}}\angle 0°}{10\angle 90°} = \frac{1}{2\sqrt{2}}\angle -90°$.

$\therefore i_{s1} = 0.5\sin(50t - 90°)$.

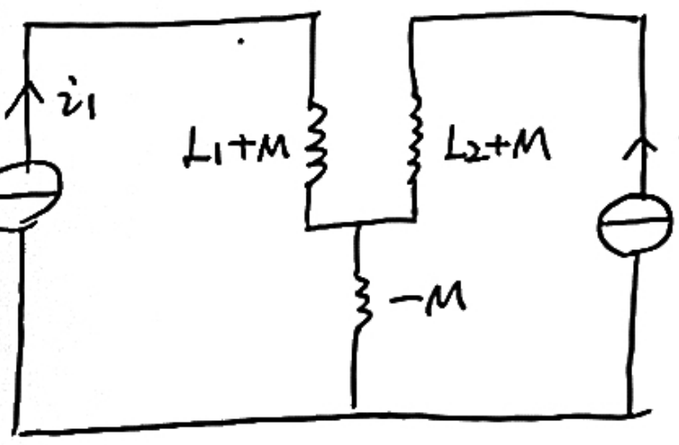
二次谐波：$w = 100$，$j\omega L = j20$，$\dot{U}_{s2} = 5\sqrt{2}\angle 0°$

$\dot{I}_{s2} = \frac{\dot{U}_{s2}}{j20} = \frac{5\sqrt{2}\angle 0°}{20\angle 90°} = \frac{\sqrt{2}}{4}\angle -90°$.

$i_{s2} = 0.5\sin(100t - 90°)$

$i(t) = [0.5\sin(50t - 90°) + 0.5\sin(100t - 90°)]$ A. **A**

15. 去耦：



$\because i_2 = 0$

$\therefore L_2 + M$ 失效.

u_2 为 $-M$ 两端电压.

$Z = -j\omega M$.

$\dot{I}_1 = \frac{I_m}{\sqrt{2}}\angle 0°$

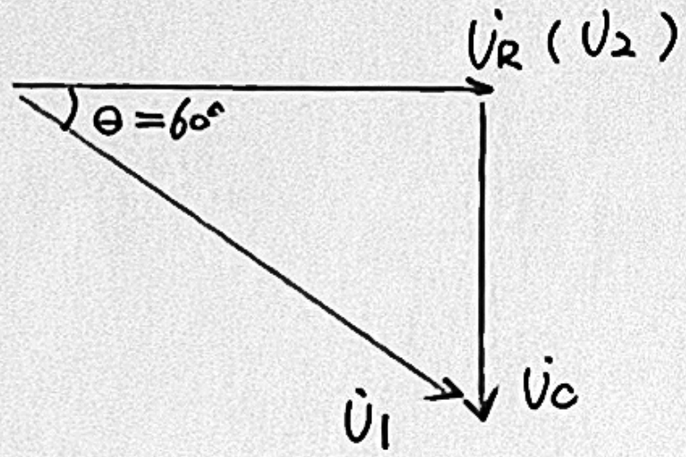
$\therefore \dot{U}_2 = \dot{I}_1 \cdot Z = \frac{I_m \cdot \omega M}{\sqrt{2}}\angle -90°$.

$\therefore u_2 = \omega M \cdot I_m \sin(\omega t - 90°)$

$= -\omega M I_m \cdot \cos \omega t$. **~~A~~B**

16 T：叠加定理，显然 D

17 T：相量图法：串联分压与串联电流为基准相量.



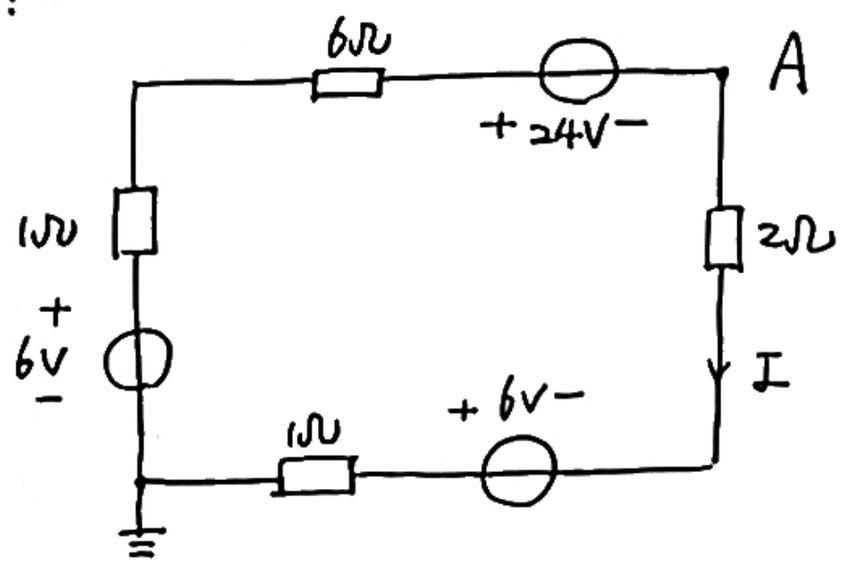
由题意：U_2 超前 U_1 60°

$\therefore \theta = 60°$

$\therefore \frac{U_2}{U_1} = \cos 60° = \frac{1}{2}$ D

18 T：$I = \sqrt{4^2 + (\frac{2.5}{\sqrt{2}})^2 + (\frac{1.5}{\sqrt{2}})^2 + (\frac{0.8}{\sqrt{2}})^2}$ C

19：(a)：



假设电流 I，顺时针 KVL：

$24 + 2I - 6 + I - 6 + I + 6I = 0$.

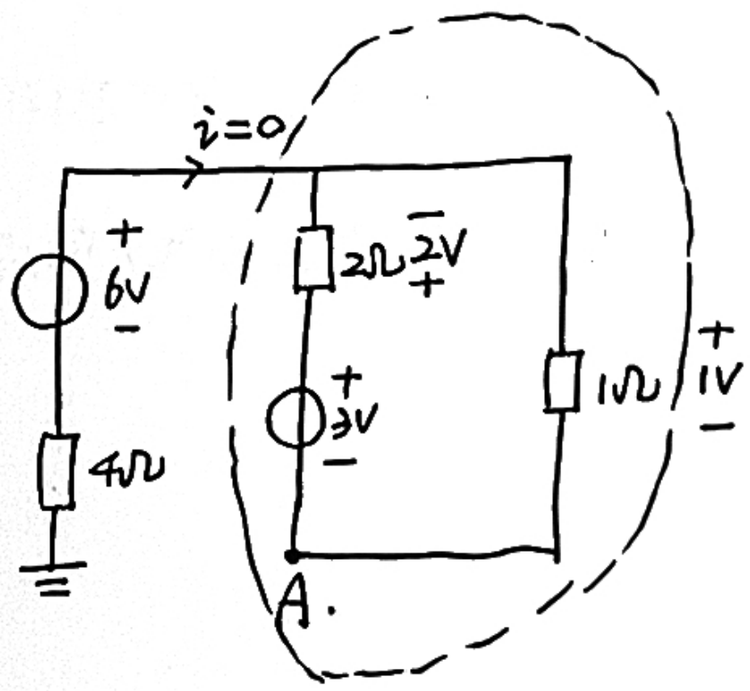
$\therefore 10I = -12$.

$I = -1.2\ A$

$\therefore \varphi_A = 2I - 6 + I$

$= 3I - 6 = -3.6 - 6 = -9.6\ V$.

(b)



分析：对虚线圈出的“广义节点”

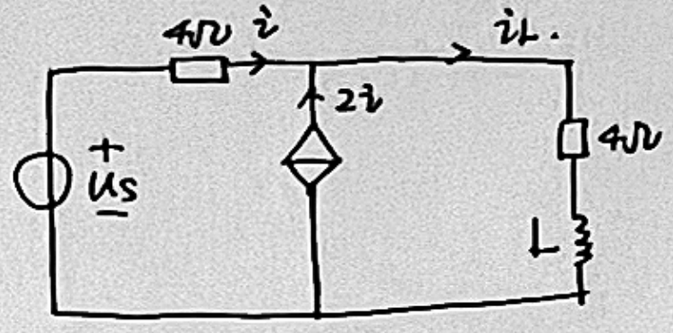
列KCL，可知 i=0，4Ω电阻短接.

故电流只在右侧回路中流.

分压可知：$U_2 = 2V$，$U_1 = 1V$

$\therefore \varphi_A = -3V + 2V + 6V = 5V$.

三、 闭合前：



$i_L = 3i$

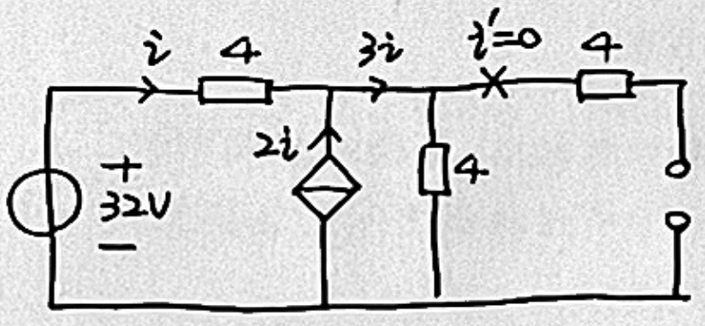
对KVL： $32 = 4i + 4i_L$.

$= 4i + 12i = 16i$

$i = 2A$ $\therefore i_L(0^-) = 6A$

$= i_L(0^+)$

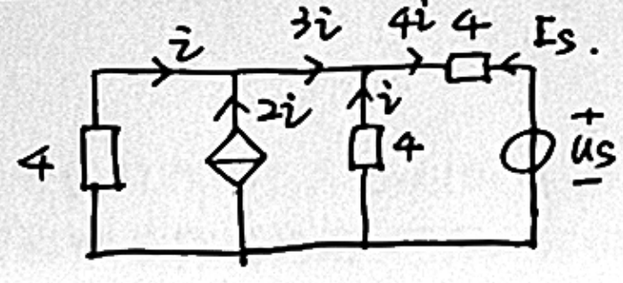
闭合后. 对L两端戴维南：



KVL： $32 = 4i + 12i = 16i$. $i = 2A$.

$\therefore U_{OC} = 12i = 24V$.

求Req. 加压求流：



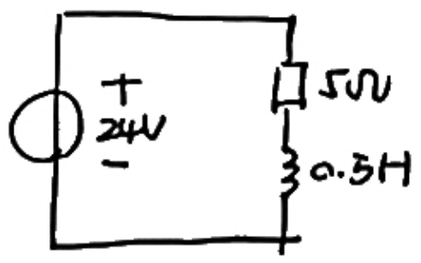
KVL： $u_S + 4i + 16i = 0$

$u_S = -20i$

$I_S = -4i$

$\therefore R_{eq} = 5\Omega$.

$\Rightarrow$



$\tau = \frac{L}{R} = 0.1s$.

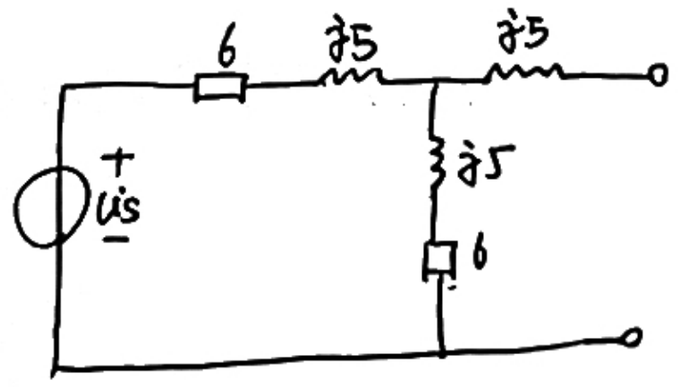
$i_L(\infty) = \frac{24}{5} = 4.8A$.

$\therefore i(t) = 4.8 + (6-4.8)e^{-10t} = 4.8 + 1.2e^{-10t}$.

$u_L(t) = L \cdot \frac{di}{dt} = \frac{1}{2} \times 1.2 \times (-10) \cdot e^{-10t} = -6e^{-10t}$.

$u_{R1}(t) = 19.2 + 4.8e^{-10t}$ $\therefore u(t) = 19.2 - 1.2e^{-10t} \cdot (V)$.

四. 去耦：

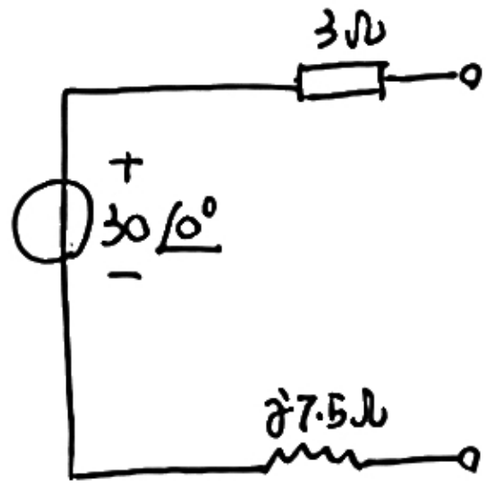


显然： $\dot{U}_{OC} = \frac{1}{2}\dot{U}_S = 30\angle 0° V$

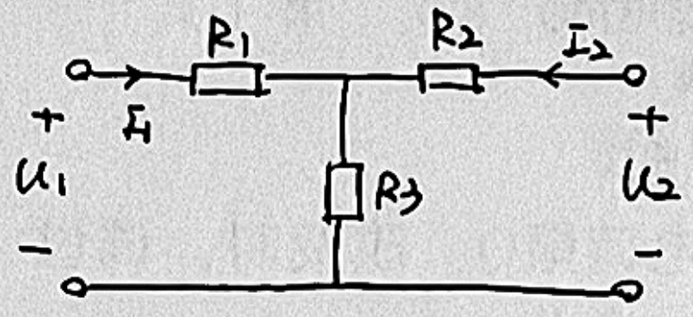
$Z_{eq} = 3 + j2.5 + j5$

$= 3 + j7.5\ \Omega$.

$\Rightarrow$



五. 已等效考虑还原电路为T型. 由于 $z_{12}=z_{21}$, 网络互易, 纯电阻.

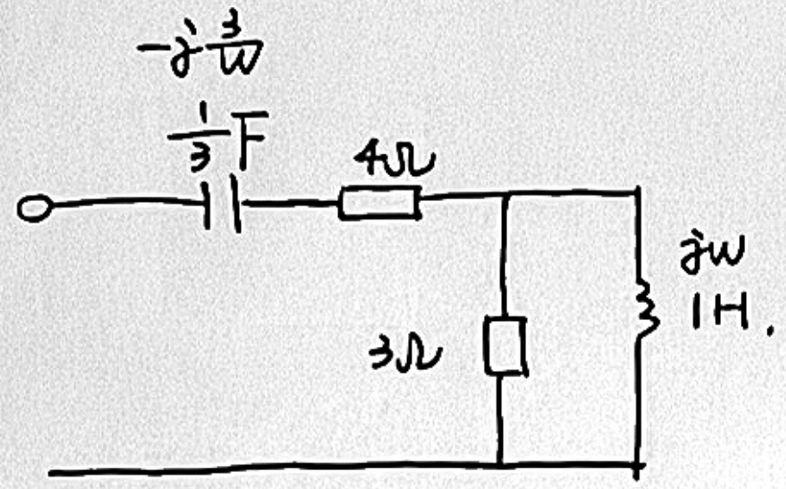


$$U_1 = I_1R_1 + R_3(I_1+I_2)$$
$$= (R_1+R_3)I_1 + R_3I_2.$$

$$U_2 = I_2R_2 + R_3(I_1+I_2)$$
$$= R_3I_1 + (R_2+R_3)I_2.$$

$\therefore R_3 = 3\Omega.$
$R_1 = 4\Omega.$
$R_2 = 0\Omega.$

$\Rightarrow$



$$Z_{eq} = \frac{j3w}{3+jw} + 4 - j\frac{3}{w}$$
$$= \frac{j3w(3-jw)}{(3+jw)(3-jw)} + 4 - j\cdot\frac{3}{w}$$
$$= \frac{j9w+3w^2}{9+w^2} + 4 - j\cdot\frac{3}{w}$$

令虚部为0: $\frac{9w}{9+w^2} = \frac{3}{w}$ $\therefore 9w^2 = 3w^2+27$ $6w^2 = 27.$

$w^2 = \frac{27}{6} = \frac{9}{2}$ $w = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$ rad/s.

(注: 一般的谐振频率 $\Leftarrow$ 令虚部为0 $\Leftarrow$ 求输入阻抗 Z_{eq}).